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Microtubules in the Heliozoan Axopodium

I. The Gradion Hypothesis of Allosterism in Structural Proteins

L. E. ROTH, D. J. PIHLAJA, AND Y. SHIGENAKA

Division of Biology, Kansas State University, Manhattan, Kansas 66502

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A heliozoan protozoan has been preserved by an improved glutaraldehyde–osmium tetroxide procedure, and the linkage structures and spacings have been described for the precise, microtubule array of the axopodium. The short link is a paired structure with a length of 2 mμ, whereas the long link is thicker with a length of 28 mμ. The axoneme cross section is composed of a triangular unit, an isosceles triangle with one 30° angle and one 29-mμ and two 55-mμ sides. The axoneme cross section is a paired series of half circles, each larger than the previous inner one by six microtubules. Anomalies that have been observed are all explicable on the basis of bonding preferences in a twelve-subunit microtubule cross section. The precision of normal ordering is also best understood on the basis of distinct subunit bonding preferences and inhibitions induced by linkage bonds. The concept of allosterism is applied; the conformational changes induced by bonding appear to act through several subunits to form an allosteric gradient. This type of gradient is termed a "gradion" and focuses attention on the externally directed bonding site, termed the "gradion site", of each microtubule subunit. The gradion hypothesis applied to other microtubule and movement systems allows new conceptual approaches to phenomena such as the variable stability, functional variability, intraspecies and interspecies variation, and formation and regulation of microtubule arrays. The concept can also be used to formulate new hypotheses of intracellular transport, cytoplasmic streaming, and axoplasmic flow.

The heliozoan axopodium is a functionally versatile and efficient structure. As a rigid body, it is probably seldom equaled; it often protrudes a quarter of a millimeter as a stiff rod into the environmental fluid but is only a few microns in diameter. Internally it exhibits a considerable amount of streaming that is seen as numerous particles moving lengthwise in both directions simultaneously. Furthermore, it can be controlled in such a way that movement of the organism is thought to be produced and that captured protozoa or rotifers can be brought to the spherical portion of the cell for ingestion. As a microtubule-containing structure, the axoneme carries out several functions in contrast to other microtubule systems that apparently carry out only one functional role.

A considerable interest has been manifest recently in the heliozoan axopodium because of these reasons and also because of its precise array of microtubules. Much of the credit for this continuing interest belongs to J. A. Kitching, who began his structural studies more than a decade ago. It was he who first demonstrated reversible axopodial retraction by hydrostatic pressure (*11*), the tubular appearance of the substituent elements of the axoneme, the "double-spiral" array of the microtubules (*12*), and the axonemal attachment to the nuclear envelope (*13*). MacDonald and Kitching (*17*) have proposed, though without giving evidence, the presence of three spacings (26, 45, and 47 mμ) produced by "links" to connect and uniformly position the microtubules, for which they also claimed to have evidence for cross sections containing twelve subunits.

Early fine-structure studies of these organisms were done by Wolfarth-Bottermann and Kruger (*40*) and Anderson and Beams (*2*).

The recent studies of Tilney have displayed improved techniques and have extended the above work. A description of the axopodium and its relation to the cell body was first presented (*33*) and was followed by experimental studies showing alterations in the microtubule array during axopodial retraction by cold (*34*), hydrostatic pressure (*35*), and colchicine (*32*) treatments.

Our own interests came from these studies, which pointed toward the similarity of this microtubule system to that of the mitotic apparatus. The ease of light microscope observation of axopodial changes allows many microtubule studies to be done quickly which would be long electron microscopic procedures if adequately carried out directly on the mitotic apparatus. This report extends the knowledge of the architecture of the axoneme which is one of the more elaborate and precise arrays of microtubules discovered. Subsequent papers deal with the subunit structure (*22*) and with further degradation and regrowth experiments on axopodia (*26*, *30*). These papers are collectively designed to present and substantiate a new and fundamental view that gives a basis for understanding the regulation and physiology of microtubule systems and furthermore directs attention to the structural proteins.

MATERIALS AND METHODS

Organisms commonly called *Actinosphaerium nucleofilum* but more appropriately and recently renamed *Echinosphaerium nucleofilum* (*9*) were obtained from the Carolina Biological Supply Company, Burlington, North Carolina and established in laboratory culture by using rotifiers and *Parameciun aurelia* as food. Considerable improvement in culture was found by using wheat infusion in pond water as the culture medium, to which a small amount of calcium carbonate was added.

Light microscope observations were carried out in chambers composed only of Pyrex glass to minimize effects from absorbed metal ions that have marked effects on axopodia

(*26*). A Zeiss photomicroscope was used with conventional Zernicke phase and Nomarski interference optics. For polarization studies a piece of Polaroid sheet was placed in the analyzer position with a standard Zeiss polarizing assembly substage.

For electron microscopy, fixation of these organisms is difficult if both the axopodia and the cell body are to be fixed well. Conventional use of glutaraldehyde and osmium tetroxide with a buffered rinse between proved entirely unsatisfactory over a wide variation of trials. The very long, slender axopodia as contrasted to the spherical cell body with numerous vacuoles (Fig. 1) present two different types of fixation conditions; the cell body was often well preserved, but altered axopodia were present (Fig. 2). Modification of typical glutaraldehyde and osmium tetroxide methods resulted in a good appearance reproducibly (Fig. 3).

With 1/30 *M* phosphate buffer at pH 5.3, 6.6, or 7.2, 2×10^{-5} *M* magnesium sulfate and 2×10^{-3} *M* sucrose included, 12% glutaraldehyde was added slowly to an equal amount of culture medium containing organisms and allowed to remain only 30 sec; the final concentrations of glutaraldehyde, phosphate buffer, magnesium sulfate, and sucrose were 6%, 1/60 *M*, 10^{-5} *M*, and 10^{-3} *M*, respectively. Excluding any rinse, osmium tetroxide fixative was added to an equal amount of the medium including glutaraldehyde fixative so that a final fixation of 20–30 min was carried out by using glutaraldehyde and osmium tetroxide together, buffered and supplemented with magnesium sulfate and sucrose; in this step, the concentrations of glutaraldehyde, osmium tetroxide, phosphate buffer, magnesium sulfate, and sucrose were 3%, 0.5%, 1/60 *M*, 10^{-5} *M*, and 10^{-3} *M*, respectively. Dehydration through 50%, 75%, and 95% ethanols was carried out for 10 min each, followed by two 10-min rinses of absolute ethanol. Propylene oxide was used in preparation for infiltration and embedding in a standard Epon–Araldite mixture.

Sections were cut by using a diamond knife and a Reichert OmU 2 microtome, mounted on substrate-free copper grids, and stained with uranyl acetate and lead citrate. Sections were observed in an RCA EMU 4 electron microscope by using the 100-kV beam and double-condenser lens system. Contamination rates as low as 3 Å/min were measured at scanning intensities of the beam without the use of cold surfaces in the specimen chamber.

OBSERVATIONS

The results described in this report are limited to a description of the architecture of the axoneme, a term used here for the microtubular array composing the major volume of each protruding axopodium (Fig. 1, arrows). Exact cross sections are by far the most useful for interpretation and can easily be obtained by sectioning from above the cell body progressively into the ectoplasmic region which includes the insertions of the axonemes into the cell body; conversely, longitudinal sections can easily be obtained by sectioning through the major diameter of the spherical cell body.

Many features of axoneme structure that were reviewed in the introduction and ascribed to previous investigators have been substantiated. Structures linking adjacent microtubules and microtubule-rows have been observed. Most frequently seen is a long-linking structure that gives an edge-to-edge spacing between microtubules of

about 28 mμ (Figs. 4–6, *L*). In some cases, this link has a diameter of about 3 mμ, but in other cases they appear thinner and show a few granules within their length (Fig. 6, *L*). In a few optimal cross sections, a given microtubule may show two such links in a V-shaped extending to the two microtubules in the next outward row (Fig. 5, *LL*). The true length of this link is probably longer than the 28 mμ, spacing that it determined because longitudinal sections show that they are not perpendicular to the microtubule axis; in the observations made, the angle is about 30–40° from perpendicular (Fig. 9, *L*) and the true link length would be about 31 mμ.

Shorter links are also seen between adjacent microtubules in the same row. The fine diameter of these links make them difficult to resolve. MacDonald and Kitching (*17*) reported these to be thicker links than the long ones but apparently did not resolve the pair. These structures are about 2 mμ long and 2 mμ in diameter and apparently can exist as paired elements between each microtubule and its neighbor in a given row although they have been observed less frequently (Figs. 5–8, *S*).

The peripheral regions of axonemes are only rarely seen to be completely filled with microtubules. Numerous missing microtubules cause open spaces in the outer sheets, particularly at or near the interfaces of the twelve sectors that comprise each axoneme (Fig. 4, arrows). The high frequency of this phenomenon suggests that it is the normal condition.

Since few microtubules extend the entire length of the axoneme and peripheral microtubules terminate with regularity along the tapering axoneme, ends of microtubules can be seen frequently in longitudinal sections. Ends typically show a dense cloud of material that is usually oriented to occupy the space at the end and toward the next, more inward microtubule (Fig. 9, *ME*).

Microtubules do not necessarily end at the nuclear envelope even though the axoneme is attached there. The major diameter of the axoneme is not at its most proximal end or at the envelope. More microtubules are present a few microns away from envelope (Fig. 9, right edge) than at the level of envelope attachment (Fig. 9, left edge). In serial cross sections, it is similarly possible to see microtubules present in one section and not in another and vice versa (Figs. 13 and 14, microtubules present denoted by *MSS*, absent ones by *X*).

Free microtubules are commonly seen almost parallel to and at some distance from axonemes both in organisms that have and have not been subjected to ex-

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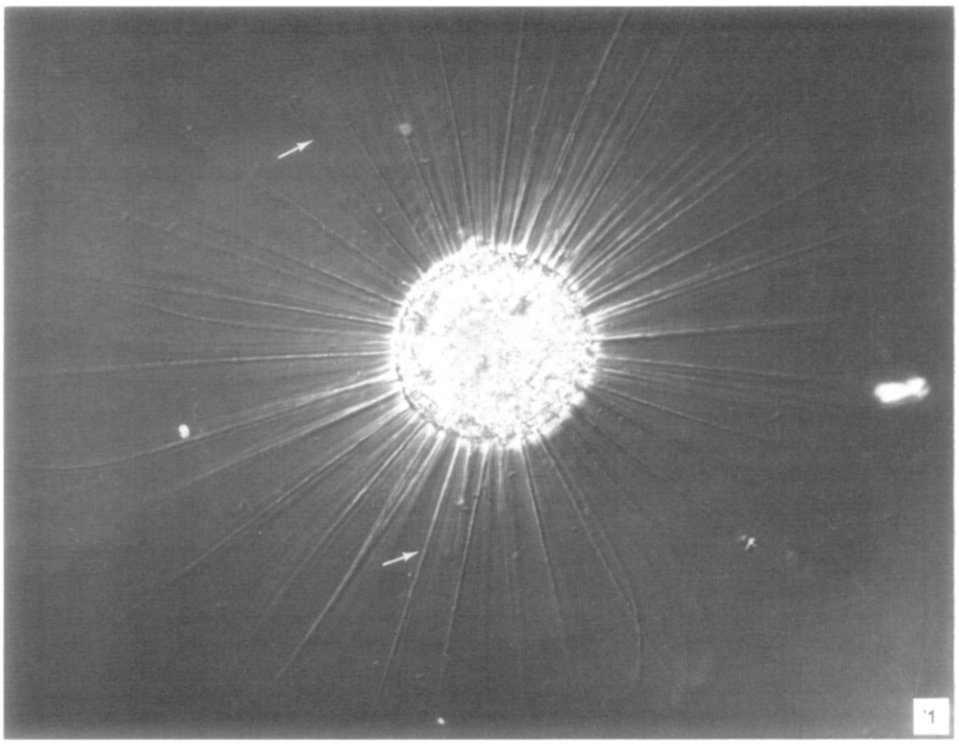
FIG. 1. A living organism showing normal axopodia (arrows) radiating from the spherical cell body. The longer axopodia approach 300 μ in length, while the cell body is about 100 μ in diameter. × 200.
FIG. 2. An embedded organism fixed in the conventional glutaraldehyde–osmium tetroxide method. Although the cell body shows some alteration, the axopodia have not been preserved. × 200.
FIG. 3. An embedded organism following fixation in the modified glutaraldehyde–osmium tetroxide method reported. × 200.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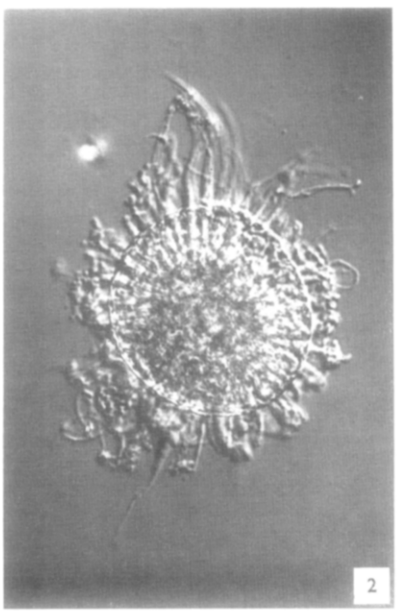
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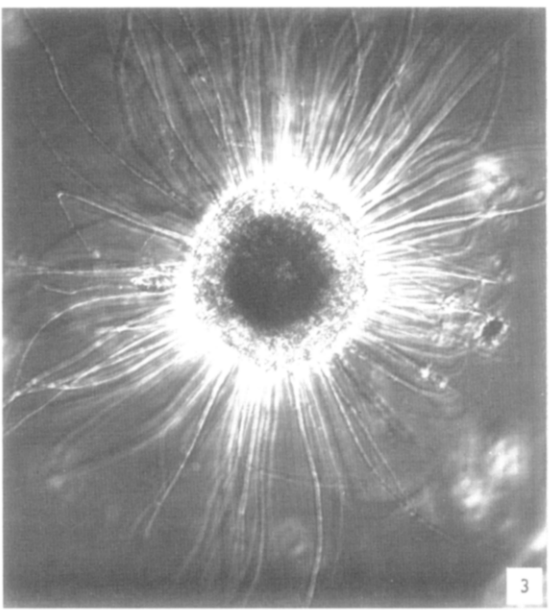
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perimental treatment designed to degrade the axonomes (Figs. 4, 10–15, and 21, *M*). Particularly in organisms that are recovering from such experimental treatments (*26*), where massive arrays of microtubules are seen, a high number of disarrayed, peripherally located microtubules are typically present (Figs. 10 and 11, *M*). Living organisms at this time show only short axopodia, e.g., 25 μ long. This experimental treatment has caused an unusually high concentration of monomer in the ectoplasmic region of the organism with the apparent result that three anomalous microtubular subarrays are formed within the normal pattern.

First, extra microtubules are frequently seen in association with the four microtubules that are considered to be the center of the eight possible locations included in this grouping of four axonemes.

Second, extra microtubules are frequently seen near the interfaces between two sectors. Typically, a single microtubule is slightly to one side of the interface (Figs. 10 and 11, *B*); however, in a few rare cases, two or three such extra microtubules together are close to one interface (Figs. 10 *BB* and 11). In extremely rare cases, a single microtubule is included between rows other than at the interfaces (Figs. 10 and 11, *BX*). On closer inspection, extra microtubules of this type have two general configurations: one can best be described as a straight Y-arrangement of four microtubules (Figs. 10 and 11, *BY*), and the other as a crooked Y with three microtubules in an equilateral-triangle arrangement (Figs. 10 and 11, *BT*).

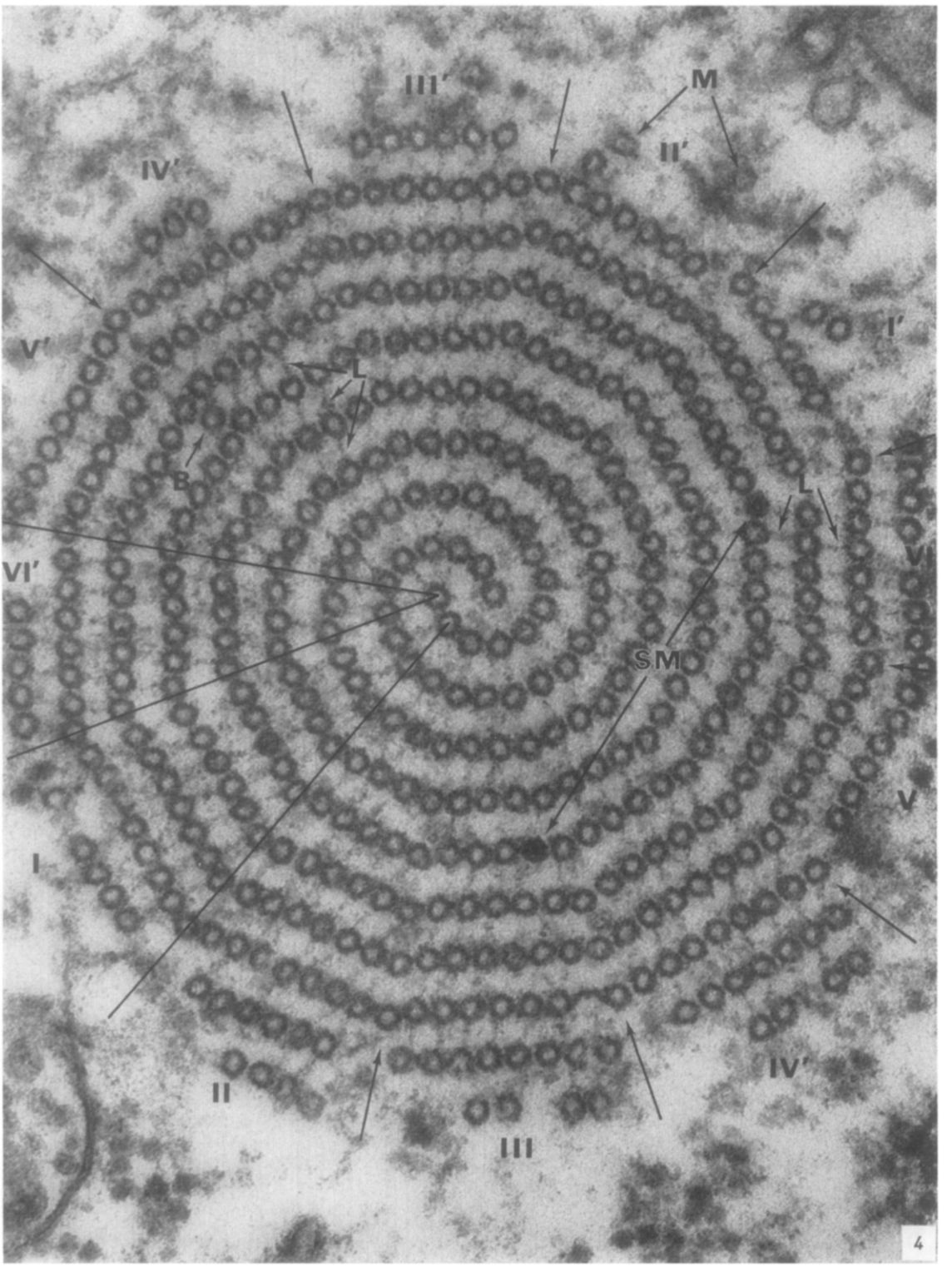
Third, peripheral anomalous arrays are present which mimic the central circular array that is normal in all axonemes. In a few cases, a close similarity to the central circular array is present (Fig. 10, *CC*), but more typically two or three new rows of microtubules seem to be growing perpendicular to the direction that they should and then bend to be parallel to the normal row arrangement (Fig. 11, *C*).

Combinations of these anomalies can be seen, particularly at the periphery of recovering axonemes (Figs. 10 and 11, *BC*).

Still more peripherally, numerous microtubules are found in random, unlinked groups or in small-group arrays, some of which are reminiscent of groupings that could be early stages of axoneme formation (Fig. 11, *AF*).

A single microtubule has never been seen surrounded by five, six or more microtubules with short-link spacings. Although four may be seen in rare cases, there is

FIG. 4. Cross section of an axoneme from a normal organism at a level inside the cell body. The twelve sectors are denoted by arrows at their interfaces. Two sectors I and I′ are slightly different than their neighboring sectors since it is there that an additional microtubule first appears in each row; each microtubule row has one additional microtubule per sector each half-turn. Sectors VI and VI′ are the last in a half turn, whereas sectors I and I′ begin a new half-turn; sectors I and I′ are based on the central microtubule parallelogram. Long links (*L*) may be seen in numerous cases, and two microtubule sections show no lumen (*SM*). One extra microtubule is present (*B*) at its typical position near a sector interface. A few microtubules not yet incorporated into the axoneme are nearby. (*M*). × 140,000.


III'
M
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V'
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L
VI'
VI
SM
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IV'

always the appearance that one of the four is too far away for a short link and too close for a long link (Figs. 10 and 11, *B*, *BB*, and *BC*). Stated conversely, microtubules have been seen with only three or less other microtubules arrayed at short-link spacings.

A single microtubule can have up to seven associated long-linked microtubules, all placed around one half of the central microtubule (Figs. 4–6). This arrangement constitutes the first and smallest microtubule half-turn and thus, normally occurs twice in each axoneme.

Axonemes that showed entirely normal spacings but with many missing microtubules were observed in normal, untreated organisms. In the cases illustrated, two small axonemes containing only two or three half-circles are shown near a more fully formed axoneme (Fig. 12). In addition, two less well formed axonemes are included (Fig. 12, upper and lower right corners). All five axonemes, even though at quite different stages of growth, have single or small groups of microtubules positioned at quite distant but still appropriate peripheral locations to start new sector-rows (Fig. 12, *MR*). Unpositioned microtubules are also present in considerable numbers (Fig. 12, *M*).

A few axonemes have been observed that lack the central parallelogram of four microtubules, at least in the plane of the section (Fig. 12, upper right corner). These are always small, loose arrays thought to be in early stages of formation; no large well-formed arrays have been seen with such a deficiency. It is suggested, however, that positioning can take place to a considerable extent without the central tetrad, e.g., to the fourth (Fig. 12, *4*) or even fifth (Fig. 12, *5*) sector-row. These arrangements are all considerably different from those seen in degrading axonemes (*26*, *30*).

Several unusual arrays have been seen and portray peculiarities observable infrequently in untreated organisms. Substitution of six microtubules for the central parallelogram array has been seen in one case only (Fig. 13, *P*). Missing groups of microtubules often replaced with vesicles are seen, although infrequently (Fig. 13, *MM*). Cleavage of large microtubule groups from axonemes is also seen, although only when the removed group has the peripheral anomally (Figs. 13–15, *C*) previously described (Figs. 10 and 11, *C* and *CC*). One case shows a large segment broken loose and another separated but not yet removed (Fig. 15, arrow). Such cleavage usually follows sector interfaces.

Two unusual microtubule cross sections have been seen. The first is a figure-eight appearance (Fig. 15, *M8*); we interpret these to be oblique sections and find no evi-

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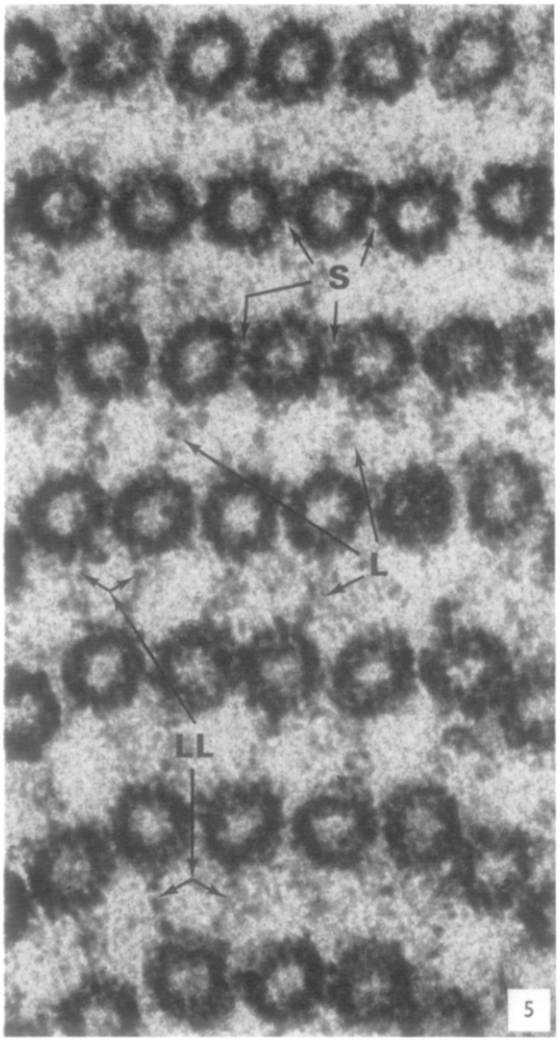
FIGS. 5–8. High magnifications of linking structures. Long links (*L*) are seen in several intrasector cases; at the sector interfaces, V-shaped, long-link pairs are present (*LL*). Some short links are included (*S*). Material in Figs. 5, 6, and 8 was fixed at pH 6.3, and that in Fig. 7 was fixed at pH 7.3. Fig. 5. ×355,000; Fig. 6. ×506,000; Fig. 7, ×216,000; Fig. 8, ×405,000.


S
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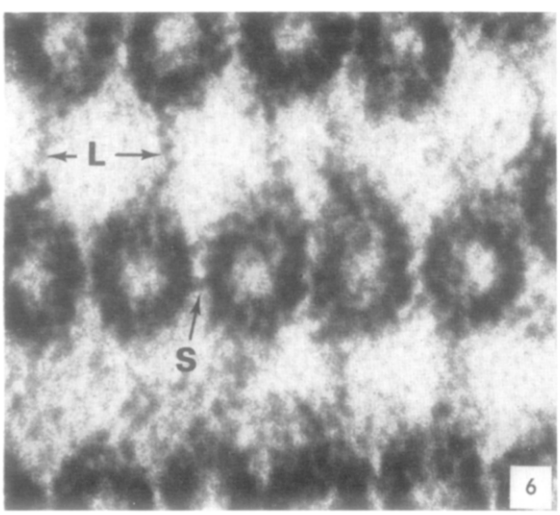
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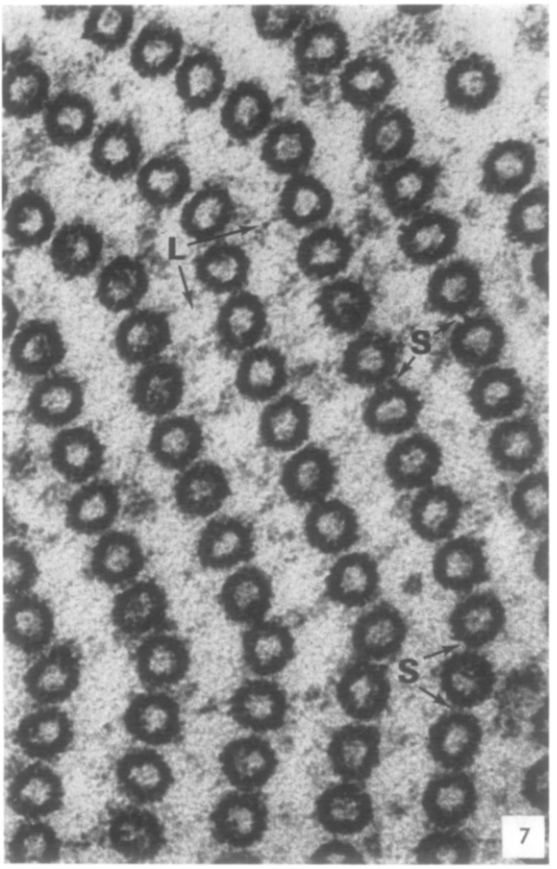
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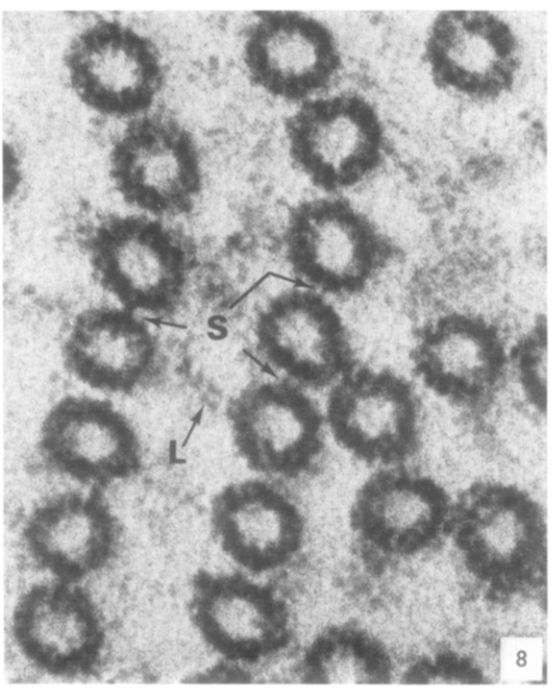
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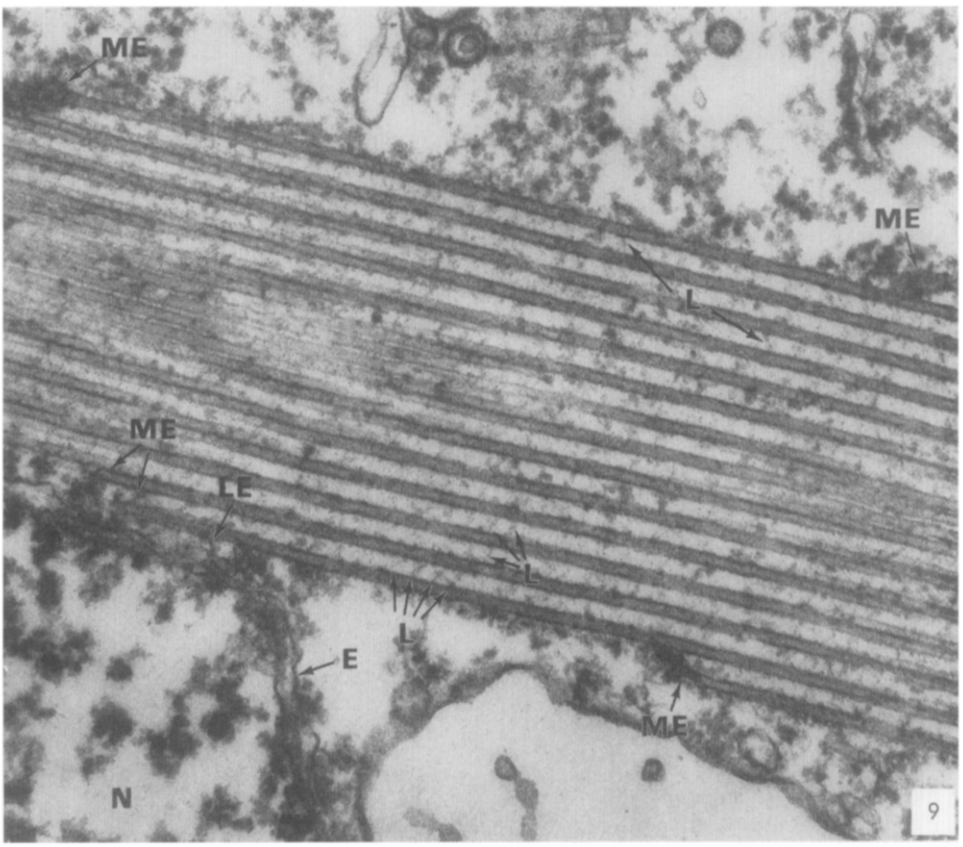
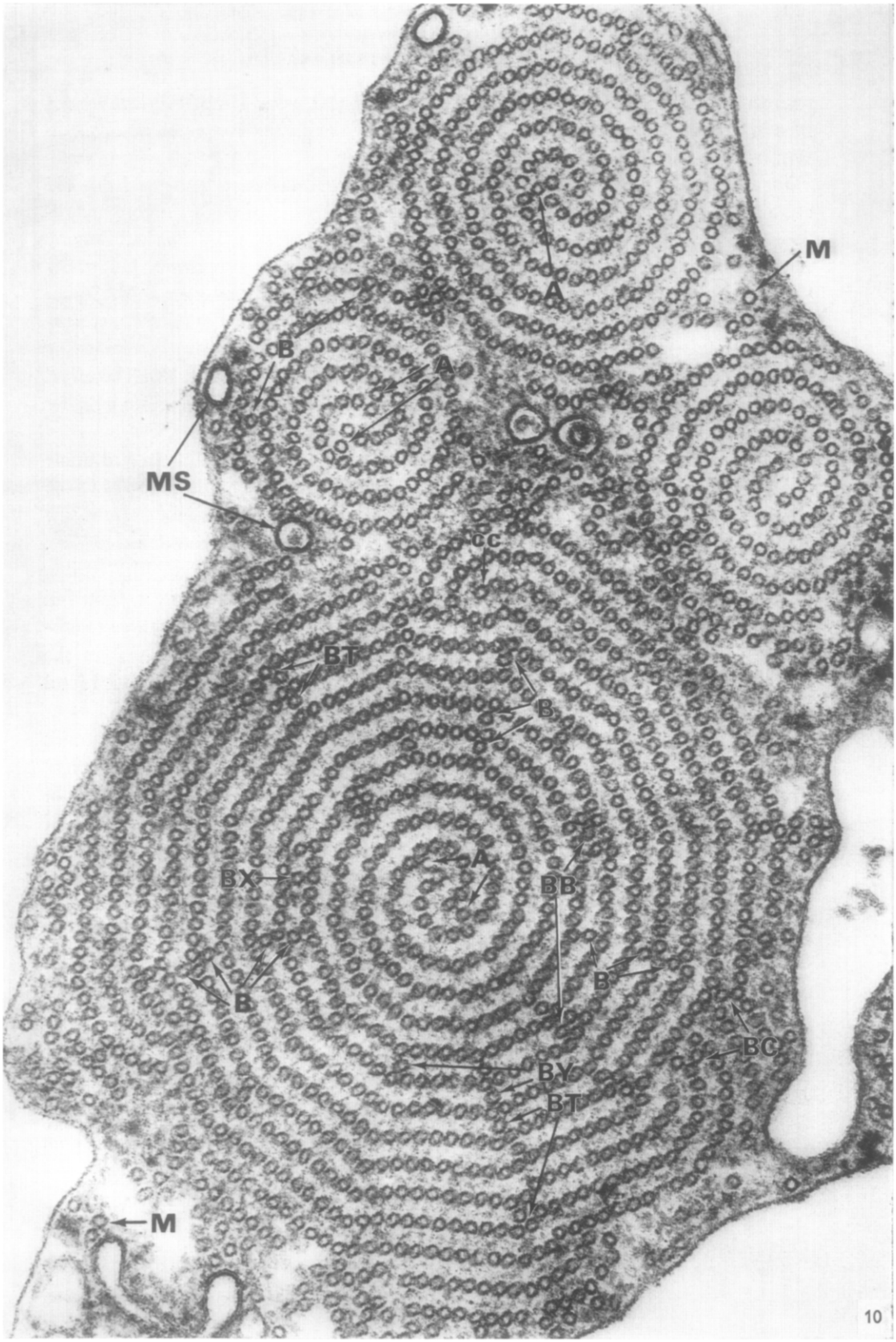


FIG. 9. Longitudinal section of axoneme near envelope (*E*) of a nucleus (*N*). Long links (*L*) are seen in several cases and are at a 30° to 40° angle to the axis of the microtubule. Several microtubule ends (*ME*) are included, and their placement demonstrates that, although the axoneme anchors on the envelope and is bound there by long links (*LE*), the major diameter is probably not at the anchoring level but slightly more distal. × 81,000.

dence that this represents a true, sectional structure. Second, lumenless microtubules have been seen on rare occasions; that is, "microtubules" are seen that are not tubular (Figs. 4 and 16, *SM*). In the later micrograph, a small axoneme sectioned near the nuclear envelope contains numerous solid cross sections along with many typical

FIG. 10. After experimental degradation (by copper sulfate treatment, see ref. *26*), reformed axonemes have anomalies probably resulting from a higher than normal concentration of subunits available first for assembly into microtubules and then into abutting, unusually numerous, abnormally short axonemes. Three anomalous, microtubule patterns have been classed into A (*A*), B (*B*, *BB*, *BY*, *BT*) and C (*CC*) and are more fully discussed in the text. Numerous unarrayed microtubules (*M*) are typically present peripheral to the axonemes. Vesicles with double membranes (*MS*) are present at locations that suggest their linkage with microtubules. × 79,000.


M
A
B
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MS
C
C
BT
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BX
A
BB
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B
BC
BY
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M
10

microtubules (Fig. 16, *SM*). The appearance in these cases is not the central dot mentioned in some observations; it is rather a fully filled circle for which no explanation is available.

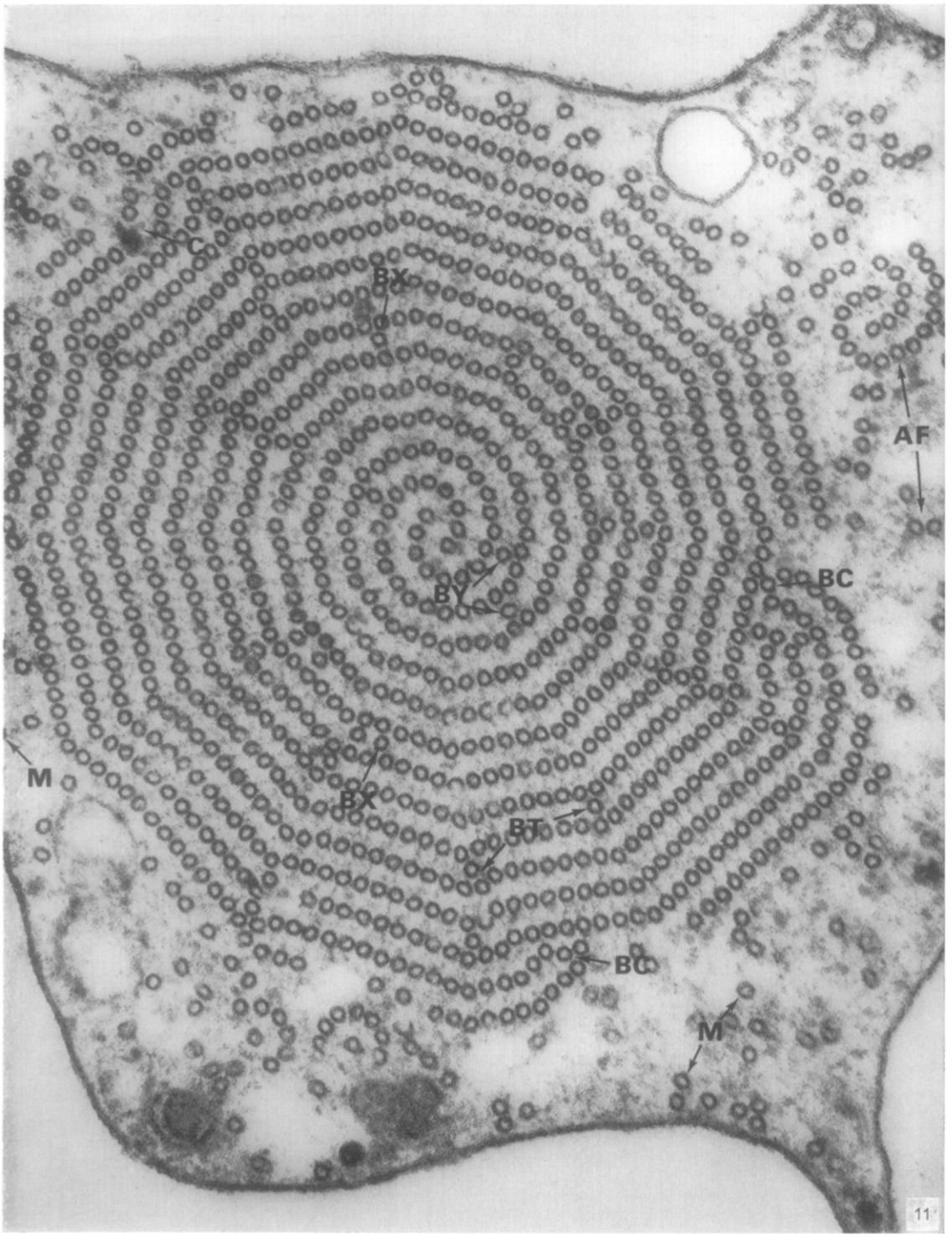
In addition to the anomalies described above, we studied organisms from old cultures in which axopodia were either short or absent. Although reduced microtubule content of such cells was expected, we found by using polarizing microscopy that brighter than normal birefringence was present in the ectoplasm. Unusually wide, birefringent, radially oriented bands were present, but they were shorter than the normal axoneme–axopodium length (Fig. 17). When such organisms were examined by using electron microscopy, large numbers of axonemes were seen including many that were relatively close together and small in diameter (Fig. 18). The effect appears to be a normal amount of microtubular mass but an aggregation into a higher number of short microtubules and axonemes.

Two analyses of the micrographs have been carried out to clarify intermicrotubule relationships. The first relates to the angle relationships between microtubules in the same row. It is quite apparent that the most common relationship is in a straight row where the angle described by the centers of three adjacent microtubules is 180°. Variations from this arrangement have been measured and tabulated. Fig. 19 is a tabulation of the angles formed by any three microtubules in the central, first, full turn of each axoneme; also in Fig. 19, the angles are recorded for the three microtubules present at the interfaces between sectors. These data suggest that in the normal axoneme in addition to the 180° placement one angular relationship is found at 150°.

The second analysis (Fig. 20) gives an indication of the frequency of short-link bonding at given subunit pairs as determined by angle relationships in both normal microtubule cases. The summary, thus obtained, portrays the high frequency of opposite orientation of short-link pairs and the progressively reduced frequency of bonding associated with subunits nearer the pair considered to have bound first.

The association of membranous structures with axonemes is so frequent that its occurrence cannot be ignored. First, vesicles with a triple-layered membrane are frequently positioned in spacings that suggest a short-link bonding to microtubules that are either free in the cytoplasm or at the periphery of axonemes (Figs. 10 and 12, *MS*). Second, small clumped vesicles with single-layered membranes are located around microtubules (Figs. 12–14 and 21, *MC*). Vesicles of both types are occasionally included in axonemes (Figs. 12–15, *MM*). Third, microtubules are located at short-

FIG. 11. Another copper-treated reforming axoneme of large diameter showing B and C type microtubule-assembly anomalies. Microtubules peripheral to the axoneme show formations that could be early axoneme centers (*AF*). Many microtubules (*M*) are located near the axoneme but as yet are not incorporated into proper spacings. × 96,000.


C
BX
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M
11

link spacings around large vesicles that contain some material slightly denser than the surrounding cytoplasm (Figs. 12–14, *MV*). Fourth, axonemes are usually anchored to the nuclear envelope and can, on rare occasions, be seen linked by the long link (Fig. 9, *LE*). Linkage of microtubules to membranes is believed to occur frequently.

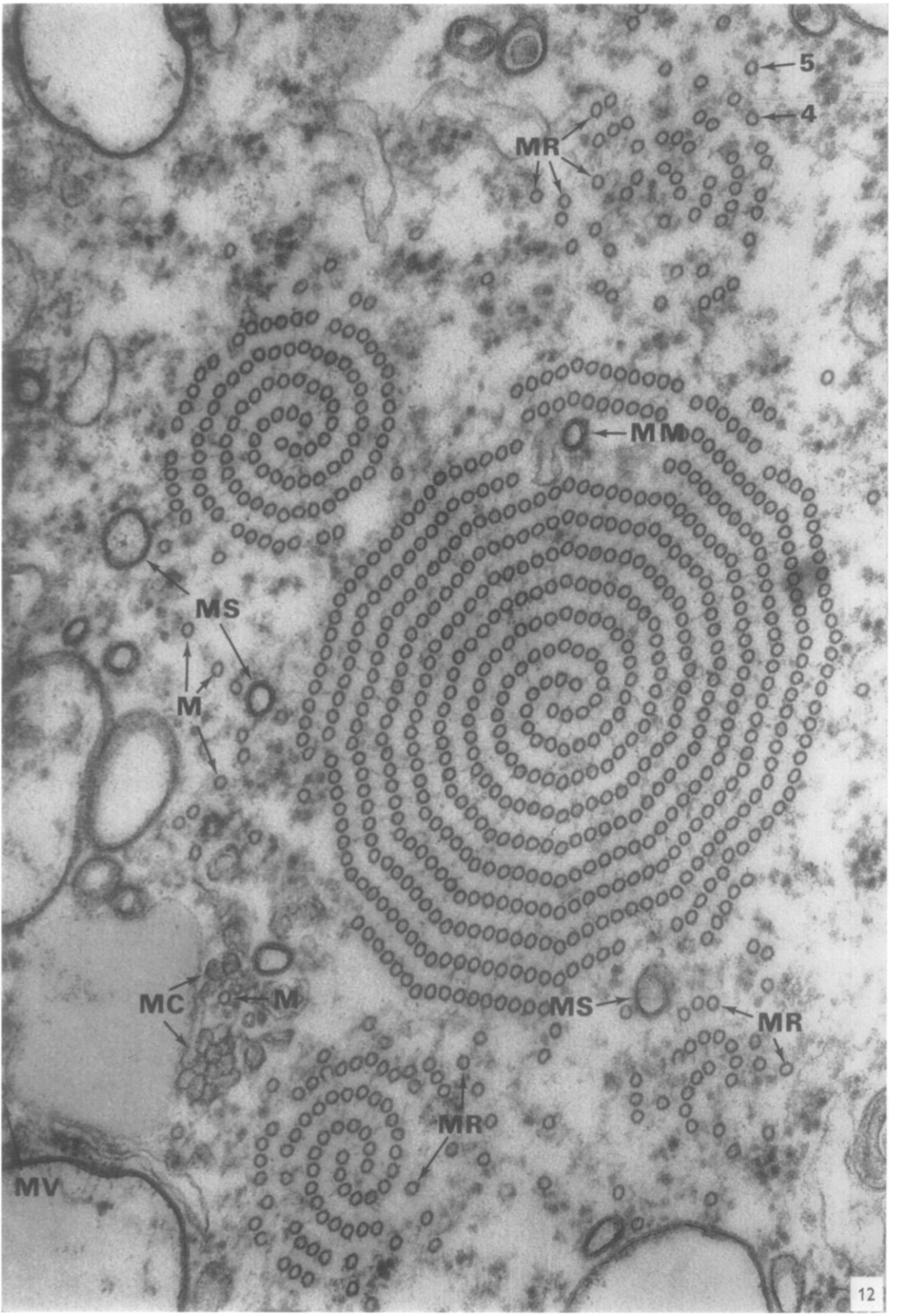
DISCUSSION

Structural analysis of the axoneme. Previous structural studies (*13*, *17*, *32*) of the heliozoan axoneme have established that a microtubule array with two or three linking structures is present as a tapering, central, axopodial element. Described as a double-spiraling array in cross section, the axoneme was shown to be degraded by cold (*34*), colchicine (*32*), and hydrostatic pressure (*11*, *35*) treatments and thus to behave quite similarly to the less organized microtubule groups in the mitotic apparatus.

Our report adds the following structural information: (1) The short spacing is or can be formed by a pair of links. (2) The long link is demonstrated to be obliquely oriented and in some cases to form a paired, V-shaped structure that connects one microtubule to two others in the next outer row. (3) A single unit triangle can describe the entire cross-sectional array. (4) An axoneme is not a true spiral (geometrically defined as a continuous curve formed by a point moving around a fixed point while steadily increasing its distance from the fixed point). An axoneme cross section is best described as a double series of nonconcentric, half circles each of which is larger than the previous inner one by the addition of six more microtubules and each of which has a radius enlarged by about one long-link spacing over the previous half circle. (5) Anomalous microtubules can be included within the usual axonemal array but preferentially in certain locations; anomalous peripheral arrays can also be formed. (6) Vesicles are frequently oriented to microtubules by spacings similar to those determined by links.

We have also collected evidence substantiated with more data than have been presented before to show that twelve subunits are found in these microtubule cross sections; these data will be presented elsewhere (*22*) but will be utilized frequently in the ensuing discussion.

FIG. 12. Five axonemes from a normal organism showing different degrees of microtubule array. Although some axonemes are incomplete, many microtubules are in their proper positions as determined by long-link spacings (*MR*) as far outward as the fourth (*4*) or fifth (*5*) sector rows. There is no indication that an axoneme is formed by microtubules being added in a single progressively extending sheet but are rather being positioned by linkage structures at any position where contact occurs. Microtubules seen as cross sections (*M*) are scattered throughout the cytoplasm. Vesicles either with double membranes (*MS*) or larger with a single membrane (*MV*) are in positions suggesting their linkage to microtubules. Other thinner-walled vesicles are also present typically in clumps (*MC*). Both vesicles may be found in axoneme faults (*MM*). × 83,000.


5
4
MR
MM
MS
M
MC
M
MS
MR
MR
MV

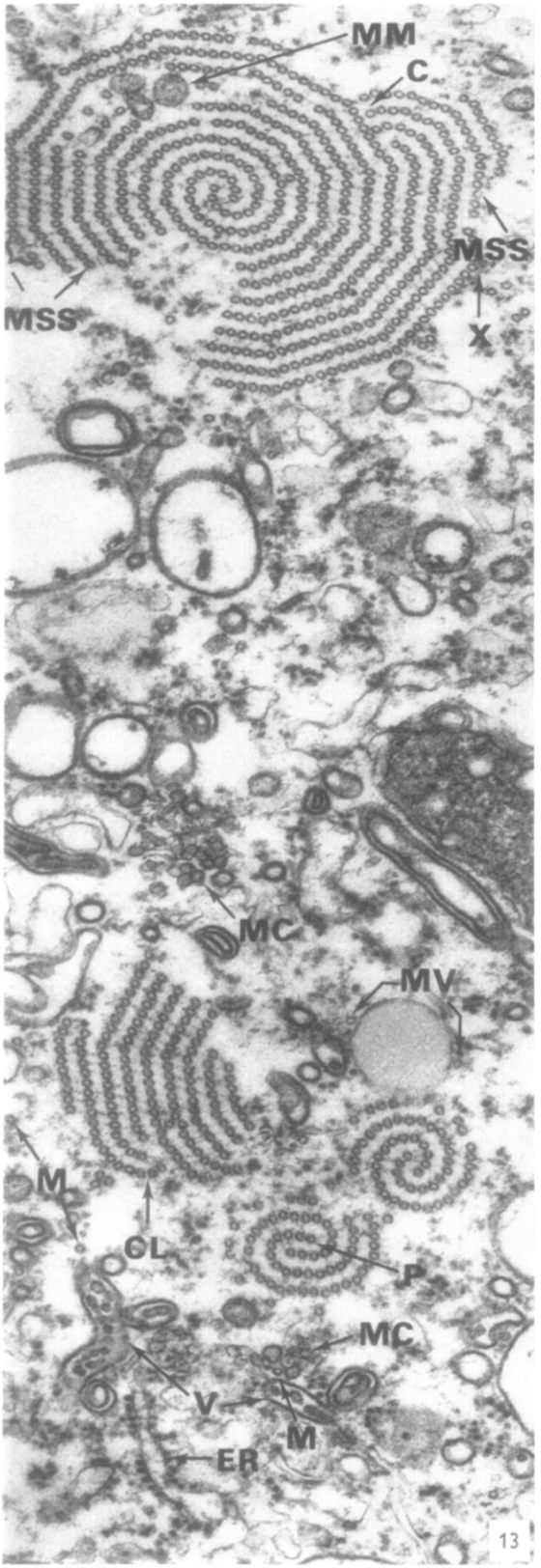
Only two measurements are important to understand the cross-sectional geometry of this preceise system; the center-to-center, short-link microtubule spacing and the center-to-center, long-link dimension. These distances can be rather accurately measured and were reported in the previous section to be 29 mμ and 55 mμ, respectively. The fundamental triangle for the entire axoneme is an isosceles triangle with one 30° and two 75° angles and one 29-mμ and two 55-mμ sides.

To check the accuracy of these measurements and the proposed triangle, trigonometric methods were used. If the proposed long spacing is used in determining the short one (assuming a 30° angle), a calculated short spacing of 28 mμ is obtained as compared to the measured 29 mμ. In another way, if the short and long ratio is used from the measurements, the calculated angle is 30° 50′ as compared to the estimated 30°. Compression in the section as a result of cutting could contribute to an error such as this, even though the axonemes chosen for such an analysis were minimally elliptical. The *ratio* of the two distances is approximately 6:11 and is, of course, more significant than the actual dimensions which are subject to the usual errors of electron microscope calibration.

Measurements for either of the links themselves cannot be stated yet with satisfying accuracy. The figures obtained by subtracting a microtubule diameter from the short-link and long-link distances are inaccurate for two reasons. The first inaccuracy comes because measurements are less accurate when made by many, single measurements of microtubule diameters as compared to measuring the sum of center-to-center distances along whole rows (as was done). Second, the angle subtended by the link and the microtubule axis in comparison to the plane of section is not known precisely enough; the few long links seen in longitudinal axoneme sections in our study suggest that the angle may be much less than 90°C. The data allow only the conclusion that a long link is about 31 mμ and the short link about 2 mμ in length, based on a microtubule diameter of 27 mμ.

FIGS. 13 and 14. Serial sections of unusual axonemal arrays. The large axoneme with one C-type array anomaly (*C*) has apparently had a group of microtubules (*CL*, Fig. 13), essentially three sectors that have two C-type anomalies, cleaved from it. In comparing the serial portions of these two micrograph, the axoneme is found to contain cases where certain microtubules (*MSS*) are contained in one but are missing from the same location (*X*) in the other sections; the locations lacking microtubules occur in both sections and indicate some nonuniformity of placement of short peripheral microtubules. A small axoneme with the central parallelogram position occupied with six microtubules has been seen in only this case (*P*). Several vesicle–microtubule associations are included: microtubules near the surface of a large vesicle (*MC*), vesicular inclusions in the axoneme (MM), clumps of small vesicles (MC), and multivesicular bodies (*V*) near microtubules (*M*) and endoplasmic reticulum (*ER*). Fig. 13, ×48,000; Fig. 14, ×60,000.

FIG. 15. An axoneme with a large group of microtubules partially broken free and another group (arrow) apparently just separating; in both cases type C anomalies (*C*) are present. An included vesicle (*MM*) and free microtubules (*M*) are present. The figure-eight microtubule appearance is interpreted as oblique sectioning (*M8*). ×40,000.



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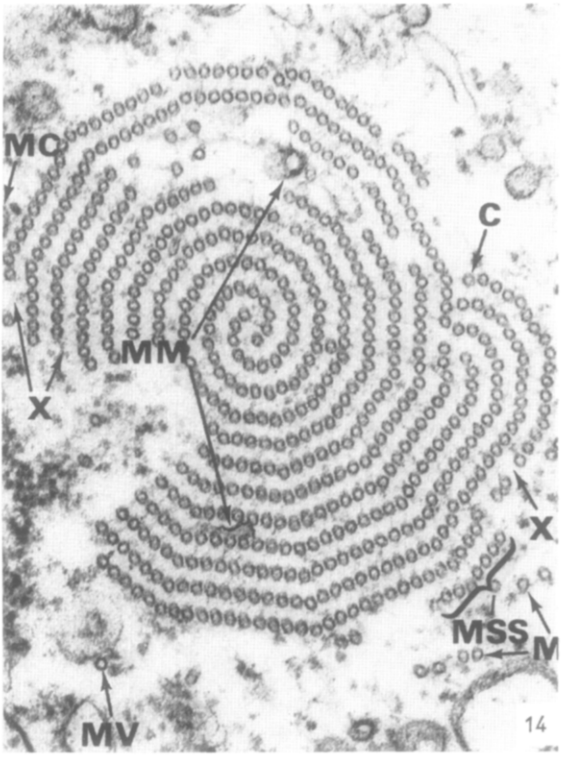

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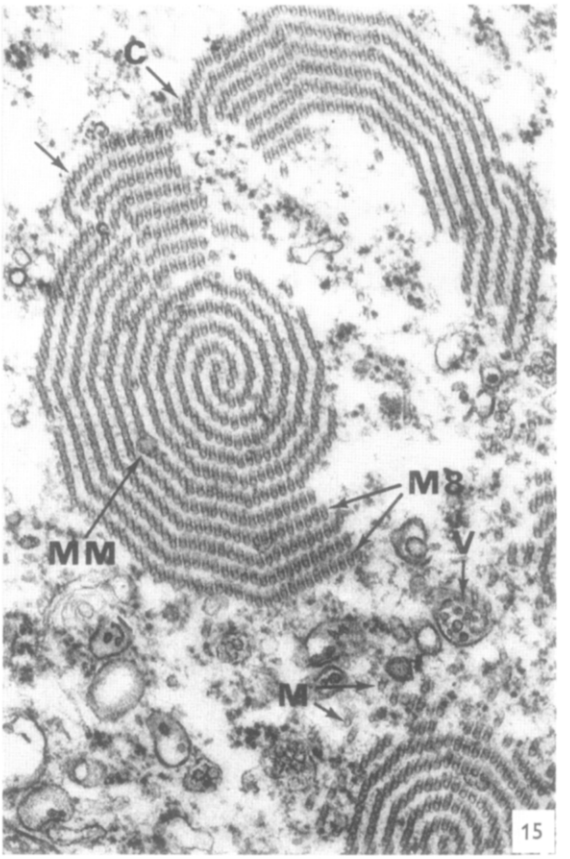

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Microtubules, it is proposed, are capable of the linkage configuration diagrammed in Fig. 22A. Four short links are attached to two subunit-pairs that are on opposite sides of the microtubule; four long links are attached similarly, two at subunit-pairs and two at subunit-pairs on the opposite side of the microtubule. Four equally spaced subunits have no attached links. One result is a basic parallelogram formed with angles 105° and 75° (Fig. 22B) and composed of two of the triangles previously described; the parallelogram is repeatedly used in this or the mirror-imaged form throughout the axoneme cross section. The sector interfaces require only that the two triangles be related in the other, slightly less favored option, and that one, short-link pair bond with a circumferential shift of one subunit to make the 150° angle discussed above (Fig. 22C).

Although such drawings are highly useful, their limitations are twofold. First, they present only a two-dimensional reconstruction of a three-dimensional system and treat the microtubule as a stacked-disk array of monomeric units when it is usually thought to be a helical array. Second, it is certainly not established by our data that each microtubule is as fully linked as described, that is eight linkage units. That sufficient structural integrity would be given by half the proposed links per microtubule is undoubtedly true, and better preservation of structure with better longitudinal views of microtubule arrays must be awaited for fuller information. Nevertheless, the linkage pattern proposed is highly useful to describe the system, to express its parameters, and to allow useful hypotheses of the *in vivo* assembly processes.

One special form of linkage involves microtubule number one in each spiral. In case only one pair of short links is present, but up to eight other links, all long ones, should be proposed; only two subunits in microtubule number one would be without links. Such a special central arrangement may be a limiting factor in the formation of new axonemes.

The concept of preferred linkage. It can hardly be coincidence that extra microtubules are so frequently found related to particular locations, that is, to the central parallelogram and the sector interfaces. This remarkable preference for extra microtubules

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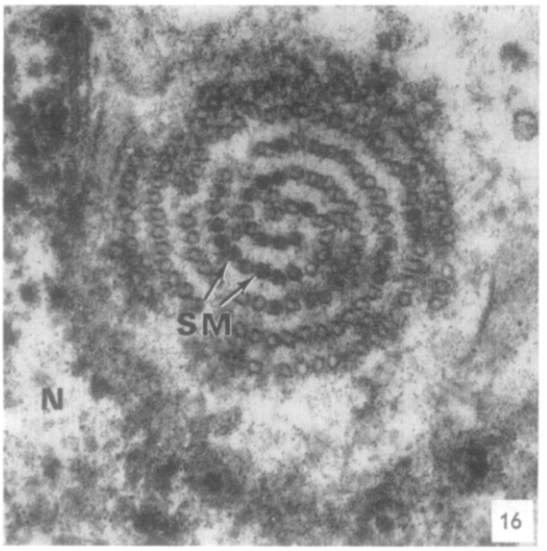
FIG. 16. An axoneme in the typical configuration in a envelope-niche of the nucleus (*N*). Cross sections of numerous microtubules appear to be filled and not have a tubular appearance (*SM*). × 68,000.

FIG. 17. Polarizing micrograph of an untreated organism from an old culture. Typically axopodia are very short, and the organisms lose their simple spherical cell-body shape. Unusually wide and bright, birefringent bands are present in the cytoplasm. × 170.

FIG. 18. Electron micrograph of the cortex of an organism from an old culture. An unusually high number of axonemes is present and are placed close together in such a way that the wide birefringent bands above can be understood as being axoneme groups. Some environmental condition has apparently induced the organism to reassemble its microtubular protein from a comparatively few, long axonemes to many short ones. × 23,000.



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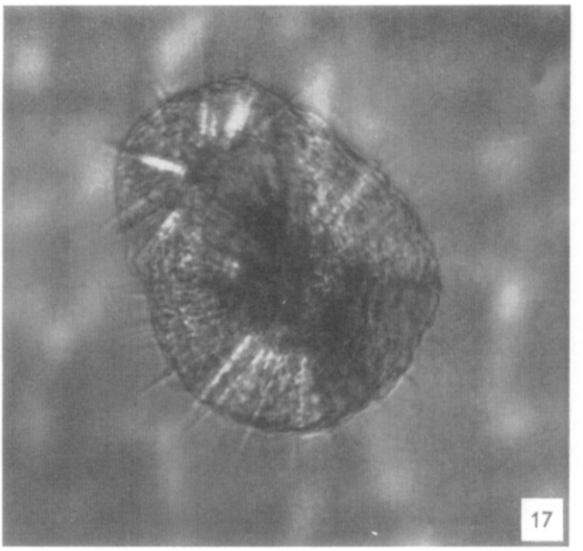

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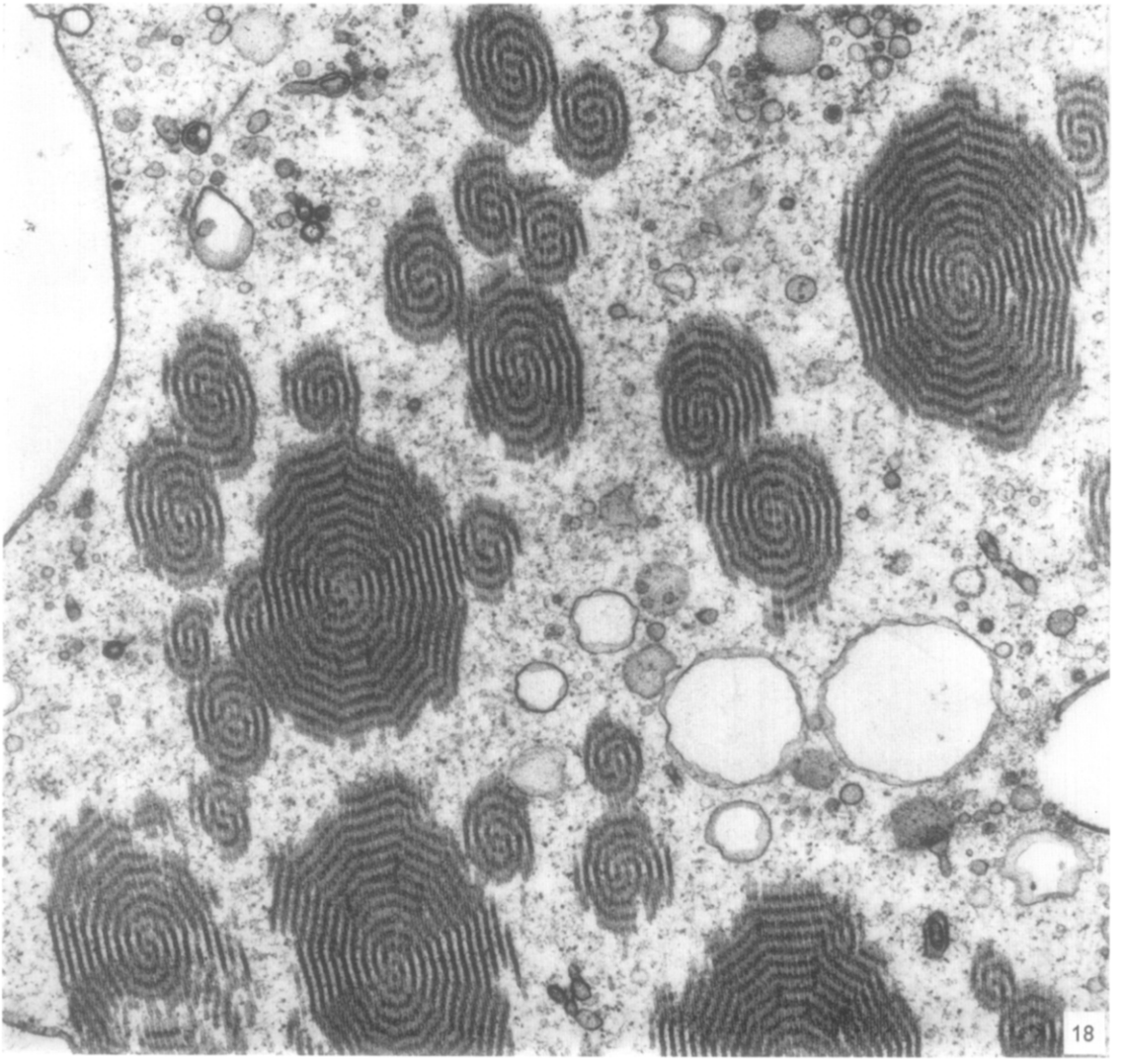

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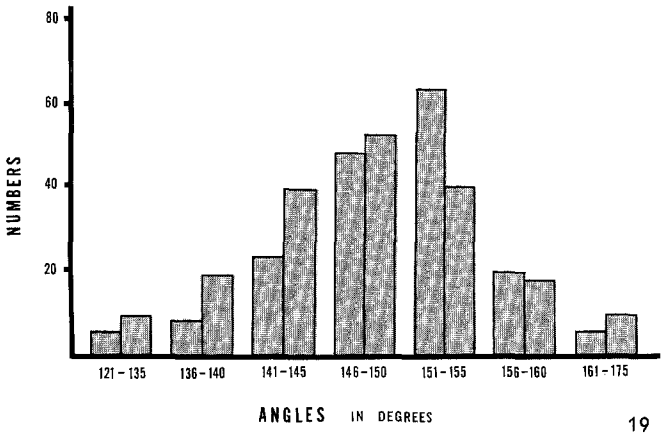

FIG. 19. Nomogram of angles formed by centers of three adjacent microtubules in the same row. The left bar in each angle group represents data from microtubule triplets in the first two half-turns in each axoneme; the right bar represents data obtained at sector interfaces that are more peripheral. Except for the 180° case, which is the most common, all other angles tend to be near 150°, according to these data.

appears to relate to microtubules that should have three or more long links bound to adjacent subunits.

Preference of bonding can be illustrated in several other cases. Most notable because it is the most common is the natural preference for pairs of both long and short links to connect at exactly opposite sides of the microtubule in the normal axoneme. Next most preferred is the 150° angle of short-link pairs with the concomitant redistribution, but still quite opposite relationships, of long links (Fig. 22C). Another example of bonding preference is seen in the peripheral arrays found in Figs. 6 and 7 (C-type anomaly); starting usually from an anomalously placed microtubule, the succeeding ones are placed in an array reflecting the tendency for half-circles.

Preference in terms of frequency of occurrence can be summarized as follows: the average, normal, axoneme cross section contains about ten sector-rows or, if completely filled, 662 microtubules, of which 540 would be in 180° linking and 122 in 150° linking. By contrast, extra microtubules as described in three classes occur only in rare or treated organisms; then each axoneme may have from 1 to 30 extra microtubules.

The statement of preference given above and summarized in Fig. 20 also implies a preference against certain other arrays. For example, even in the numerous microtubules seen peripheral to an axoneme, no hexagonal packing of six microtubules around a central one has been observed in this organism; not even four or five with proper short spacings have been seen. *In regard to the number of short links attached to*

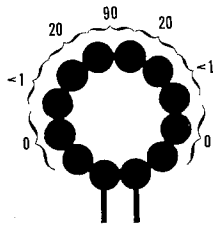

FIG. 20. Summary of occurrence of short-link bonding when one pair of short links is present. Including both normal and abnormal axonemes, the most likely case is bonding to subunits exactly opposite from the other pair, whereas the next most frequent is the 150° case at the sector interface. Other cases are extremely rare; in fact, of the three possibilities that remain, only one has been seen to occur. The occupancy ratios noted are approximate but suggest the graded series of occupancy frequency that suggests the gradion concept.

one microtubule, two pairs are preferred, one and three are acceptable, and four through six are unacceptable.

Another expected but not observed pattern is the long single-layered sheet of microtubules frequently seen in protozoa and other cells. Except as included in the axonemal rows, such sheets of microtubules are not seen and seem to be prohibited as free arrays in the cytoplasm. Such restrictions imply that, following the bonding of one link, preference is given for the opposite link to be attached next.

The following summary of preferences best fits the available data.

1. The binding of either type of link causes a preference for the next such link to be attached to the opposite subunits. That is, if a pair of short links binds to subunits 1 and 2, the next, preferred short-link subunits are numbers 7 and 8; if a long link binds to number 1, the next preferred, long-link position is number 7.

2. The early binding of either type of link causes a preference for the next link to be of the *other* type and to be attached to the closest possible subunit. That is, if a pair of short links binds to subunits 1 and 2, the next preferred, long-link position is number 4. (Although subunit 3 is the closest, it is probably unusable simply because of geometric restrictions.) Similarly, if a long-link binds to subunit number 1, the next short-link positions are subunits 3 and 4.

The formation of an axoneme. The following considerations are based on the assumptions that both synthesis of proteins and assembly of microtubules are separable in time and location from axoneme assembly. This concept has been discussed generally in a previous publication (*23*) and is a useful approach to the consideration of organelle formation.

A small axoneme that is essentially the central 12 to 16 microtubules could form by several possible sequences, no one of which is likely to be exclusively used. Logical-

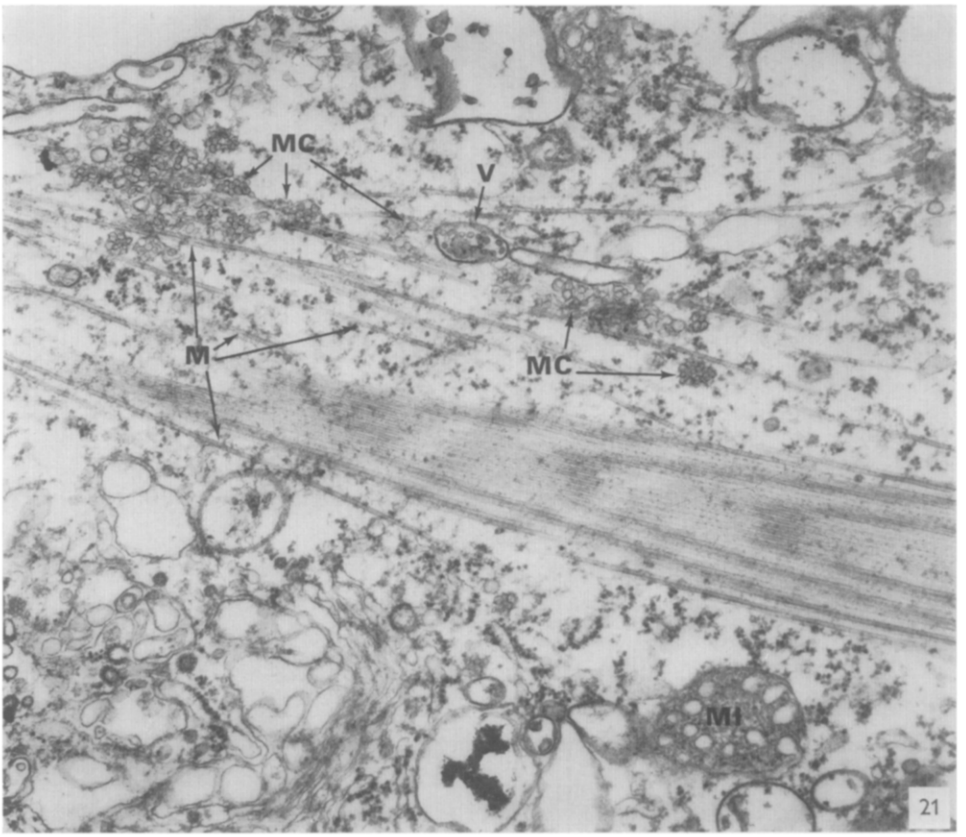

FIG. 21. Longitudinal section of an axoneme showing large clumps of small vesicles (*MC*) around some of the free microtubules (*M*) near the axoneme. A multivesicular body is present nearby (*V*). The lower part of the micrograph includes a highly vesicular region that may be Golgi material and a mitochondrion (*MI*). ×30,000.

ly, a central parallelogram could form with subsequent microtubules being added sequentially; small assemblies indicative of such a beginning have been seen, but other patterns have also been found. In further assembly, a microtubule must "choose" between the possibility of 180° or 150° short-link angles. Thus, in the second half-turn, one 180°-linked microtubule is placed between two 150°-linked microtubules; in each succeeding half-turn, an additional 180°-linked microtubule is included to each sector-row. The sector facets, thus, begin to appear in the second half-turn. Since typical large axonemes almost always show incomplete sector facets at their periphery, it is probable that the 180° linkage "choice" preferentially takes

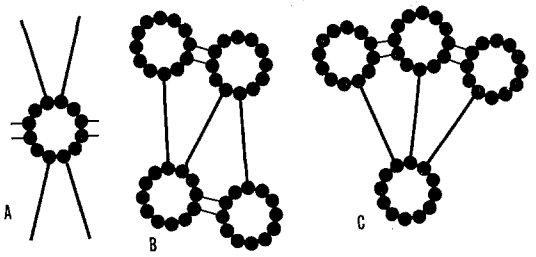

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FIG. 22. Interpretive drawings of microtubules and their linkage pattern in *Echinosphaerium* axonemes. Part A depicts the most typical, maximum, linkage capability of a microtubule. Part B shows the typical parallelogram arrangement that is the center of the axoneme and is repeated throughout the structure. Part C portrays the microtubule-link pattern at a sector interface. In both B and C, the pattern is composed of two isosceles triangles whose apices are microtubule centers. Such triangles whose equal sides are quite close to two times the length of the base is the unit structure that can be used to describe the cross-sectional pattern of the entire axonemal array.

place beginning from a mid-facet location and proceeds until a niche is produced into which a microtubule may fall if it will take the 150° orientation of short links.

We concur with Tilney (*31*) that short microtubules are formed at a different location and then move, usually one by one, into their proper linkage locations near the base of the axoneme. Such an assumption is based on the fact that in normal organisms individual microtubules are regularly seen in parallel orientations close to the axoneme, especially in organisms recovering from copper treatment; here the monomer concentration is presumed to be very high. (10^9 subunits could have been suddenly added to the cytoplasmic pool as a result of axoneme degradation.) In such cases, two or more microtubules commonly come together away from the axoneme in the configurations discussed by Tilney (*31*); extra microtubules within the axoneme could result from the insertion of such preformed groups of two or three microtubules. An alternate and preferred explanation of these inclusions is that the interfaces of sectors poses a unique situation in which the usual (long-link) preference is altered. The latter explanation is favored because of the rarity of anomalous microtubular inclusions elsewhere in the axonemal bundle.

But more germane to our discussion, let axoneme formation be considered at the molecular level and be based on a more molecularly oriented description of the entire microtubule system. An average-sized axoneme of ten sector-rows would have 662 microtubules at its major diameter (55 microtubules per sector plus two central ones). Each organism has about 100 axopodia that measure about 300 μ from base to tip; since central microtubules are probably full length and more peripheral ones progressively shorter, average microtubule length can be estimated at 150 μ. Total microtubule length then equals 1.5×10^2 μ (average microtubule length) times 10^2 (number

of axonemes) times 6.6×10^2 (microtubules/axoneme) or 10^7 μ per organism; laid end to end, the microtubules in one organism are estimated to extend about ten meters. Each micron can be estimated to have 25 turns of 12 subunits each, or 300 subunits; without including any residual cytoplasmic monomer pool, each organism would have about 3×10^9 subunits in its axonemes.

Long links, at their highest frequency, occur paired about five times per micron of microtubule length; they would number about 10^8 in the collected axonemes of an organism. Short links, assuming the maximum of one pair per turn per microtubule, would number about 50 per micron and total 5×10^8. The ratios of long links to short links to microtubule monomer would be approximately 1:5:30.

Although subject to several comparatively minor corrections, this ratio points out the preponderance of microtubule subunits and the comparatively low number of the largest of the three elements, the long link. The cell certainly must produce much more microtubule monomer than links.

It must be considered an absolute necessity that random bonding of links with free monomer be excluded since it would spell disaster for the formation of an ordered structure. If so allowed, the high amount of subunit monomer would, by concepts of mass action, be expected to bond to most free links and then be rather randomly incorporated into forming microtubules according to where the subunit (with attached link) appeared to contact and bond; links are and could not be randomly placed to form this highly ordered pattern. Conversely, polymerization of monomer must be strongly encouraged. Therefore, in terms of the linkage site, the subunit must be considered to have two forms, one that is inactive and, after polymerization, one that is activated to bond with linkage elements.

The following four features are proposed as the rate-regulators for axoneme growth. First, nucleation is needed to produce the first turn of new microtubules. Second, synthesis of linkage elements could limit axoneme growth in both the addition of new microtubules and elongation of the assembly. Third, the microclimate (pH, ions) and availability of building blocks at microtubule tips would determine the direction of the polymerization or depolymerization. Fourth, allosteric restraints as discussed above would rate-limit the speed of growth.

Obviously, more mechanisms must be invoked to explain the constraints and precision of the axoneme formation process than have been hitherto suggested in structural systems. What properties of proteins can be utilized to explain such a precise self-assembly process?

The application of allosterism. The postulation of two forms of the microtubule protein and the concept of preferred linkage discussed above refer to highly specific, organized, molecular interactions. They imply (1) indirect interactions between different binding sites, (2) multiple, interacting binding sites, and (3) interactions that

may be either cooperative or antagonistic. All three features are cardinal to the phenomenon of allosterism in enzymatic proteins, a model of which was presented by Monod *et al.* (*19*).

The data now available on enzymatic proteins lead to valuable new understanding and postulation when applied to structural proteins such as those in microtubules. It is a primary contention of this paper that *structural proteins are capable of allosteric interactions and must be understood within the framework of allosterism as now applied to enzymatic proteins.* This concept calls attention to the multifunctional nature of a single protein species; a given primary structure or amino acid sequence can exist in several secondary or tertiary conformations that have different capabilities. Or, stated differently, the colinearity of DNA and protein describes not one function but a series or range of molecular shapes and bonding niches; a protein should be considered changeable, flexible, or variable within a functional range that is specified and limited by its amino acid sequence.

Furthermore, the conformational state of the molecule is regulated by small molecules that cause controllable changes in molecular shape. The regulatory capabilities of such a mechanism are immense but have been little applied to structural studies or to considerations of cellular function.

Two cases may be cited as examples of physiological utilization of the concept. The first is an attempt to consider the function of the neuronal membrane, a subject recently reviewed by Lehninger (*16*). He cited and expanded on the postulation of Mueller and Rudin (*20*) that a group of five or six alamethicin molecules may aggregate in a stack extending through the membrane, block K^+ transport in one state, and allow or participate in transport in another state that is induced by an electrical stimulus. The second pertains to invertebrate respiratory proteins (*41*).

The phenomena described for the constraints of microtubule assembly in the heliozoan axoneme are best explained on the basis of allosterism. This study serves as a first case where evidence is available to point out the usefulness of allosterism applied to cell structure.

In this context, activation of a free binding site following subunit polymerization is immediately understandable on the basis that the subunit protein undergoes a conformational modification when incorporated into the microtubule. In terms of enzymology, inhibitor or activator molecules can bond to oligomers but not to monomers; in terms of microtubules, links may bond to microtubules but not to free microtubule monomer. No radical shape change is necessarily involved since it is established that a protein can go from one state to another and still conserve its symmetry in a polymer (*19*).

Major alterations of protein conformation, however, have been shown to destroy the action of effector molecules. That is, treatments by urea, temperature change, or

pH change eliminate the effects of small, effector molecules so that the structure entirely breaks down. The effector molecule may remain attached to the oligomer so that reformation by removal of treatment may proceed more quickly than original assembly.

Allosteric mechanisms are greatly favored and regulation is more easily effected if the functional state of the protein employs two or more identical protomers in an oligomer. Thus, it is probable that the microtubule subunit may be an oligomer of one or more monomers. Furthermore, evidence is now being presented that a protein may contain one subunit, amino acid chain for each stereospecific binding site (*8*). In terms of microtubules, the subunit is probably at least a dimer. Data (*7*) interpreted as indicating binding of one colchicine molecule with a microtubule protein dimer have previously been given. Furthermore, to account for the specificity of the sites involved in polymerization to form a microtubule, two or three specific amino acid chains (or subunits, using this word to denote a segment of the "subunit" denoted elsewhere in this paper) should probably be postulated.

These concepts are fruitful in providing new hypotheses to explain microtubule arrays and functions. The two cases most valuable are, first, the variable stability of different microtubule systems and, second, the genetic control of various microtubule arrays.

Much has been written about the *varying stability and lability of microtubule systems*. The assumption has been made that this effect is most likely to be due do to different protein subunits (*4*, *23*). However, evidence is available from studies of glutamine synthetase (*10*, *29*) that some conformational states are more refractory to experimental degradation than others; for example, the taut form is resistant to disaggregation of protomers by organic mercurials and urea at pH 8.0, whereas the relaxed form is not. In terms of microtubules, it is necessary to consider not only the protein per se, but also the type of links bound at surface sites and the degree of occupancy of these sites; these factors must greatly affect the reaction to experimental degradation and the functional competence of microtubules. Further evidence for the application of this concept to a microtubular system will be given in one of our subsequent papers (*30*).

For example, microtubules in cilia, where many cross-connecting structural elements exist, are much more resistant to experimental degradation than are microtubules in the mitotic apparatus. Similarly, microtubules in early division stages of the mitotic apparatus are thought to be more labile than are the same microtubules in the telophase midbody. Again, cases have been noted where more stable microtubules are closely associated with membrane surfaces; included are the tendencies for microtubules to form there (*23*, *25*) and to be last seen there in experimental degradation (*24*). Certainly linkage to any cell structure or even merely occupancy by monovalent

ions without linkage could confer such stability to microtubules by causing a change in conformation of the protein subunits.

In vitro studies of microtubule proteins are being attempted currently and should utilize these concepts to advantage. Stable conformations of the isolated protein should be experimentally induced and manipulated to the advantage of the researcher without the fear that the protein is actually degraded whenever it is more stable than in the *in vivo* state. The formation of crystals of microtubules following vincristine sulfate treatment (*5*) should thus be interpreted as a highly bound, stabilized state of the protein; if properly altered it can be induced to "decrystallize" and then polymerize into microtubules *in vitro;* the microtubule state could be favored by the addition of compounds to occupy surface sites or to cross-link the surface sites of adjacent microtubules.

Structural variation and its genetic determination in microtubule systems is also newly approached by these considerations. The formation of the existing variety of systems must be considered as a self-assembly process which should be extended to include large structures, such as the axoneme, which is almost macroscopic. These structural assemblies should be considered as subject to variation on the basis of the available building blocks such as linkage structures and the natural constraints to their assembly. Mass-action phenomena are implicated in the kinetics, and a certain amount of error would be expected, e.g., extra microtubules in axonemes.

Genetic control of structure can be exerted not only by what microtubule monomer supplied but also what molecules are produced to occupy surface sites and cross-link the microtubules. The different assemblages reported, for example by Tucker (*36*), could be explained as well by the presence of a new linkage molecule being produced as it could by suggesting that a different microtubule protein exists.

Production of small activating molecules can also be invoked to control the time when the assembly process begins, the rate at which it proceeds, and the extent to which it goes. It is entirely possible to postulate much more strongly than previously reasonable that only one or a few microtubule monomers are made in all cells and are interchangeably used in cilia, the mitotic apparatus, centrioles, and other microtubule systems, and that their differing functions are a result of varying regulatory and linking ions and molecules.

In summary, structural and physiological variation are explicable in more than one way. To invoke a different protein for each variation is unnecessary and, at times, is entirely unsatisfactory. Variation within a given individual microtubule assembly is such a case; see Tucker (*36*) regarding structural change and the work on urea effects on axonomes (*30*).

THE GRADION HYPOTHESIS

In considering the protein chemistry of enzymes, one difficulty in the above structural applications is that the enzymes studied thus far are usually active as dimers, tetramers, and hexamers (or as such assemblies paired) in contrast to the essentially infinite-polymer that a microtubule should be considered. In enzymology it has tended to be sufficient to consider allosterism only as it relates to the subunit adjacent to a changed subunit. Furthermore, the enzymologist has often thought in terms of only two conformational states.

Protein considerations need to be extended to explain the microtubule phenomena presented here. In the axoneme, we have evidence that allosterism can be transmitted across several subunits; the binding at subunits one and two of a pair of short links affects the bonding at number four; furthermore, in perhaps a graded way, bonding at subunits six and seven is also affected. Since we lack direct evidence from enzyme studies for such a distance of action, what feasibility questions can be directed to the enzymologist when we need to postulate a graded, allosteric effect across as many as six subunits?

First, in contrast to the two conformational states usually considered for most *in vitro* enzyme studies, can several states exist for one oligomer? The answer is certainly affirmative. The study of glutamine synthetase by Shapiro and Ginsburg deals with the interconversion of three well-known states of the double-hexamer components (*29*). They propose a slow, two-stage conformational change from the relaxed to the tightened form, indicating that at least four states are possible, some of which are dependent in part on divalent cations. Further, Koshland, Nemethy, and Filmer proposed in 1966 (*15*) a sequential model based on the principle that a ligand can induce a conformational change in a given subunit which may change the conformational stability of neighboring subunits *in varying degrees*. Later studies (*14*) on glyceraldehyde-3-phosphate dehydrogenase were interpreted to show that this enzyme "appears to fit a sequential model." Clearly multiple states or perhaps a continuum of many conformational states are possible with a maximum number as yet undetermined.

We propose therefore that microtubules be understood on the basis of a graded, sequential effect over numerous subunits. The unit of variation thus created in an otherwise infinite polymer is termed "the gradion". A gradion is defined as a functional unit consisting of a linear series of subunits each of which is slightly different in conformation from its neighbors; the "end" subunits will then present one extreme of bonding capability with the other extreme being midway in the gradion. The "gradion site" is that subunit bonding site available to links on the microtubule surface.

Such a gradion considered in a direction essentially parallel to the microtubule length could explain the regular but infrequent spacing of long links in the axoneme. After bonding a long link, allosteric changes induced in neighboring subunits would preclude another such linkage until the gradient inhibition was reduced by enough subunits to be weak or absent. After bonding, the new link would exert its gradion effect in both directions so that no long link would be allowed between it and the previous link. The indicated gradion length in this case is perhaps five to seven subunits.

A circumferential gradion could explain most of the constraints and preferences reported in this paper. The maximum gradion would be about ten subunits in this case, and the diagram previously presented (Fig. 20) can essentially be construed as a description of the frequency of bonding within this gradion. It is significant that such a gradion needs to utilize only about four different subunit states, the number recently suggested by enzyme studies (*14*, *29*).

Intracellular transport is frequently associated with microtubule assemblies: axopodial streaming (*1*), melanin-granule transport (*6*), food intake by suctorian tentacles (*27*), lens formation in the squid eye (*3*), and axoplasmic flow in neurons (*37*).

That membrane-bound vesicles are associated with, and probably linked to, microtubules is demonstrated in this report. An attraction and tendency to bond can be suggested to exist between membrane surfaces and the microtubule surfaces that have free gradion sites. Such sites are potentially useful in various functional capacities and can be considered as key points in an hypothesis of transport involving stationary microtubules.

In the heliozoan axoneme, such free sites would be considered in gradions formed by linkages on the attached surfaces of the microtubule and by allosteric effects transmitted circumferentially. Linear gradions would be indicated by the frequency of long links, and the gradion length could be considered as essentially the same as the long-link, repeat distance. In considering transport of vesicles, a steady-state condition can be postulated so that a vesicle is bonded to each gradion; all vesicles would be subjected to equal gradions on either side. If a vesicle is removed at any point, an unequal gradion condition results, and the vesicle is induced to move to a new equilibrium position. Continuous flow requires continuous removal of vesicles at one end, plus an energy input to provide for bond formation and/or breakage. A rolling action of vesicle or protein globule is likely and could cause a transmission of movement to other materials somewhat spatially removed from the gradion itself and help to create a wider stream of moving material.

Schmitt (*28*) has suggested a concept of transport in nerve axons that has rudiments of the gradion concept. However, he has not utilized the allosterism concept but alludes only to protein-protein bonding in regard to microtubule–vesicle relationships.

It is difficult to see how simple bonding can give rise to such a highly dynamic phenomenon as axoplasmic flow nearly as well as would the gradion concept.

Such particulate transport could be invoked also in the absence of microtubules. In cytoplasmic streaming, where studies have usually shown a fine linearly arranged material (*39*) and only rarely any evidence for microtubule arrays (*18*), fibrous sheets are frequently described and could function similarly.

Intracellular movements could also be given fresh conceptual approaches by application of the gradion concept. In the mitotic apparatus, microtubules are apparently implicated in chromosome movement; the first definitive observations of links (*38*) are just now being made and add data to the quite frequent observation for groupings of about a dozen microtubules into fibrils. In neurons, microtubules are being implicated in axoplasmic flow, and linkage structures are just now being observed, particularly in the axon hillock and initial segment (*21*).

In cilia and flagella, the gradion hypothesis presents the possibility of a transmitted alteration in conformation and size of subunits and focuses on the importance of the surface-associated structures on microtubules as the place where a change may cause movement-initiating conformational changes in the microtubular elements. The ready association of membranes with microtubules extends the considerations to membranes and suggests that the structural proteins of membranes be similarly considered. Furthermore, it is possible to consider heterogeneous gradions consisting of more than one protein or molecular species and including more than one organelle as a means of intraorganelle or even intracellular linkage and regulation.

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REFERENCES

1. ALLEN, R. D., *in* ALLEN, R. D. and KAMIYA, N. (Eds.), Primitive Motile Systems in Cell Biology, p. 407. Academic Press, New York, 1964.
2. ANDERSON, E. E. and BEAMS, H. W., *J. Protozool.* **7**, 190 (1960).
3. ARNOLD, J. A., *J. Ultrastructure Res.* **17**, 527 (1967).
4. BEHNKE, O. and FÖRER, A., *J. Cell Sci.* **2**, 169 (1967).
5. BENSCH, K. G. and MALAWISTA, S. E., *J. Cell Biol.* **40**, 95 (1969).
6. BIKLE, D., TILNEY, L. G. and PORTER, K. R., *Protoplasma* **61**, 322 (1966).
7. BORISY, G. G. and TAYLOR, E. W., *J. Cell Biol.* **34**, 525 (1967).
8. GERHART, J. C. and SCHACHMAN, H. K., *Biochemistry* **4**, 1054 (1965).
9. HOVASSE, P., *Protistologica* **1**, 81 (1965).
10. KINGDOM, H. S., HUBBARD, J. S. and STADTMAN, E. R., *Biochemistry* **7**, 2136 (1968).
11. KITCHING, J. A., *J. Exptl. Biol.* **34**, 511 (1957).

12. —— *in* ALLEN, R. D. and KAMIYA, N. (Eds.), Primitive Motile Systems in Cell Biology, p. 445. Academic Press, New York, 1964.
13. KITCHING, J. A. and CRAGGS, S., *Exptl Cell Res.* **40**, 658 (1965).
14. KOSHLAND, D. E., JR., CONWAY, A. and KIRTLEY, M. E., *in* Regulation of Enzyme Activity and Allosteric Interactions, p. 131 (Federation European Biochem. Soc., 4th meeting, Vol. 1). Academic Press, New York, 1968.
15. KOSHLAND, D. E., JR., NEMETHY, G. and FILMER, D., *Biochemistry* **5**, 365 (1966).
16. LEHNINGER, A. L., *Proc. Natl. Acad. Sci. U.S.* **60**, 1069 (1968).
17. MACDONALD, A. C. and KITCHING, J. A., *Nature* **215**, 99 (1967).
18. MCMANUS, M. A. and ROTH, L. E., *J. Ultrastruct. Res.* **20**, 260 (1967).
19. MONOD, J., WYMAN, J. and CHANGEUX, J.-P., *J. Mol. Biol.* **12**, 88 (1965).
20. MUELLER, P. and RUDIN, D. O., *Nature* **217**, 713 (1968).
21. PALAY, S. L., SOTELO, C., PETERS, A. and ORKAND, P. M., *J. Cell Biol.* **38**, 193 (1968).
22. PIHLAJA, D. J., ROTH, L. E. and SHIGENAKA, Y., in preparation.
23. ROTH, L. E., *in* ALLEN, R. D. and KAMIYA, N. (Eds.), Primitive Motile Systems in Cell Biology, p. 527, Academic Press, New York, 1964.
24. ROTH, L. E., *J. Cell Biol.* **34**, 47 (1967).
25. ROTH, L. E. and SHIGENAKA, Y., *J. Cell Biol.* **20**, 249 (1964).
26. ROTH, L. E., SHIGENAKA, Y. and PIHLAJA, D. J., in preparation.
27. RUDZINSKA, M. A., *Trans. N.Y. Acad. Sci., Ser. II*, **29**, 512 (1967).
28. SCHMITT, F. O., *Proc. Natl. Acad. Sci. U.S.* **60**, 1092 (1968).
29. SHAPIRO, B. M. and GINSBURG, A., *Biochemistry* **7**, 2153 (1968).
30. SHIGENAKA, Y., ROTH, L. E. and PIHLAJA, D. J., in preparation.
31. TILNEY, L. G. *in* The Emergence of Order in Developing Systems, *Develop. Biol.*, Suppl. 2, p. 63.
32. —— *J. Cell Sci.* **3**, 549 (1968).
33. TILNEY, L. G. and PORTER, K. R., *Protoplasma* **60**, 317 (1965).
34. —— *J. Cell Biol.* **34**, 327 (1967).
35. TILNEY, L. G., HIRAMOTO, Y. and MARSLANG, D., *J. Cell Biol.* **29**, 77 (1966).
36. TUCKER, J. B., *J. Cell Sci.* **3**, 493 (1968).
37. WEISS, P., *Neurosci. Res. Program Bull.* **5**, 371 (1967).
38. WILSON, H. J., *J. Cell Biol.* **40**, 854 (1969).
39. WOHLFARTH-BOTTERMANN, K. E., *in* ALLEN, R. D. and KAMIYA, N. (Eds.), Primitive Motile Systems in Cell Biology, p. 79. Academic Press, New York, 1964.
40. WOHLFARTH-BOTTERMANN, K. E. and KRUGER, F., *Protoplasma* **43**, 177 (1964).
41. WYMAN, J., *J. Mol. Biol.* **39**, 523 (1969).